

81. Let $a_1 = b_1$ and $a_n = a_{n-1} + (n-1)$, $b_n = b_{n-1} + a_{n-1}$, $\forall n \geq 2$. If $S = \sum_{n=1}^{10} \frac{b_n}{2^n}$ and $T = \sum_{n=1}^8 \frac{n}{2^{n-1}}$, then $2^7(2S - T)$ is equal to _____.

माना $a_1 = b_1$ तथा $a_n = a_{n-1} + (n-1)$, $b_n = b_{n-1} + a_{n-1}$, $\forall n \geq 2$ है। यदि $S = \sum_{n=1}^{10} \frac{b_n}{2^n}$ तथा $T = \sum_{n=1}^8 \frac{n}{2^{n-1}}$ हैं, तो $2^7(2S - T)$ बराबर है।

Ans. 461

Sol. $S = \frac{b_1}{2} + \frac{b_2}{2^2} + \frac{b_3}{2^3} + \dots + \frac{b_{10}}{2^{10}} \quad (A)$

$$\frac{S}{2} = \frac{b_1}{2^2} + \frac{b_2}{2^3} + \frac{b_3}{2^4} + \dots + \frac{b_{10}}{2^{11}} \quad (B)$$

$$(A) - (B) \Rightarrow \frac{S}{2} = \frac{b_1}{2} + \frac{a_1}{2^2} + \frac{a_2}{2^3} + \dots + \frac{a_9}{2^{10}} - \frac{b_{10}}{2^{11}}$$

$$\Rightarrow S = b_1 + \frac{a_1}{2} + \frac{a_2}{2^2} + \dots + \frac{a_9}{2^9} - \frac{b_{10}}{2^{10}} \quad (C)$$

$$\Rightarrow \frac{S}{2} = \frac{b_1}{2} + \frac{a_1}{2^2} + \dots + \frac{a_9}{2^{10}} - \frac{b_{10}}{2^{11}} \quad (D)$$

$$(B) - (D) \Rightarrow \frac{S}{2} = \frac{b_1}{2} + \frac{a_1}{2} + \frac{1}{2^2} + \frac{2}{2^3} + \dots + \frac{8}{2^9} - \frac{a_9}{2^{10}} - \frac{b_{10}}{2^{11}}$$

$$\Rightarrow \frac{S}{2} = \frac{1}{2} + \frac{1}{2} + \frac{1}{2^2} \left(\frac{1}{1} + \frac{2}{2} + \dots + \frac{8}{2^7} \right) - \frac{a_9}{2^{10}} - \frac{b_{10}}{2^{11}}$$

$$\Rightarrow \frac{S}{2} = 1 + \frac{1}{4}(T) - \frac{a_9}{2^{10}} - \frac{b_{10}}{2^{11}}$$

$$\Rightarrow 2S = 4 + T - \frac{a_9}{2^8} - \frac{b_{10}}{2^9}$$

$$\Rightarrow 2S - T = 4 - \frac{a_9}{2^8} - \frac{b_{10}}{2^9}$$

$$\Rightarrow 2^7(2S - T) = 2^9 - \frac{37}{2} - \frac{130}{2^2}$$

$$= 461$$

82. Let $a_1, a_2, \dots, a_n$ be in A.P. If $a_5 = 2a_7$ and $a_{11} = 18$, then

$$12 \left(\frac{1}{\sqrt{a_{10}} + \sqrt{a_{11}}} + \frac{1}{\sqrt{a_{11}} + \sqrt{a_{12}}} + \dots + \frac{1}{\sqrt{a_{17}} + \sqrt{a_{18}}} \right) \text{ is equal to } \underline{\hspace{2cm}}.$$

माना $a_1, a_2, \dots, a_n$ A.P में है। यदि $a_5 = 2a_7$ तथा $a_{11} = 18$ है, तो

$$12 \left(\frac{1}{\sqrt{a_{10}} + \sqrt{a_{11}}} + \frac{1}{\sqrt{a_{11}} + \sqrt{a_{12}}} + \dots + \frac{1}{\sqrt{a_{17}} + \sqrt{a_{18}}} \right) \text{ बराबर है।}$$

Ans. 8

Sol. Given $a + 4d = 2(a + 6d)$

$$a + 8d = 0 \dots\dots\dots (1)$$

$$\text{and } a_{11} = 18 \Rightarrow a + 10d = 18 \dots\dots\dots (2)$$

From (1) and (2)

$$-8d + 10d = 18 \Rightarrow d = 9 \text{ and } a = -72$$

$$a_{10} = a + 9d = -72 + 81 = 9$$

$$a_{18} = a + 17d = -72 + 153 = 81$$

Now

$$12 \left[\frac{1}{\sqrt{a_{10}} + \sqrt{a_{11}}} + \frac{1}{\sqrt{a_{11}} + \sqrt{a_{12}}} + \dots\dots\dots + \frac{1}{\sqrt{a_{17}} + \sqrt{a_{18}}} \right]$$

$$12 \left[\frac{\sqrt{a_{10}} - \sqrt{a_{11}}}{-d} + \frac{\sqrt{a_{11}} - \sqrt{a_{12}}}{-d} + \dots\dots\dots + \frac{\sqrt{a_{17}} - \sqrt{a_{18}}}{-d} \right]$$

$$12 \left[\frac{\sqrt{a_{10}} - \sqrt{a_{18}}}{-d} \right] = 12 \left[\frac{3-9}{-9} \right] = 12 \times \frac{6}{9} = 8$$

83. Number of 4-digit numbers that are less than or equal to 2800 and either divisible by 3 or by 11, is equal to _____.

4-अंकों की संख्याओं, जो 2800 से कम या इसके बराबर है तथा 3 या 11 से विभाज्य है, की संख्या है।

Ans. 710

Sol. 1000 – 2799

Divisible by 3

$$1002 + (n - 1)3 = 2799$$

$$\mathbf{n = 600}$$

Divisible by 11

$$1 - 2799 \rightarrow \left[\frac{2799}{11} \right] = [254] = 254$$

$$1 - 999 = \left[\frac{999}{11} \right] = 90$$

$$1000 - 2799 = 254 - 90 = 164$$

Divisible by 33

$$1 - 2799 \rightarrow \left[\frac{2799}{33} \right] = 84$$

$$1 - 999 \rightarrow \left[\frac{999}{33} \right] = 30$$

$$1000 - 2799 \rightarrow 54$$

$$\therefore n(3) + n(11) - n(33)$$

$$600 + 164 - 54 = 710$$

84. The sum $1^2 - 2 \cdot 3^2 + 3 \cdot 5^2 - 4 \cdot 7^2 + 5 \cdot 9^2 - \dots + 15 \cdot 29^2$ is _____.

$1^2 - 2 \cdot 3^2 + 3 \cdot 5^2 - 4 \cdot 7^2 + 5 \cdot 9^2 - \dots + 15 \cdot 29^2$ बराबर है।

Ans. 6952

Sol. Separating odd placed and even placed terms we get

$$S = (1 \cdot 1^2 + 3 \cdot 5^2 + \dots + 15 \cdot (29)^2) - (2 \cdot 3^2 + 4 \cdot 7^2 + \dots + 14 \cdot (27)^2)$$

$$S = \sum_{n=1}^8 (2n-1)(4n-3)^2 - \sum_{n=1}^7 (2n)(4n-1)^2$$

$$\text{Applying summation formula we get} = 29856 - 22904 = 6952$$

85. Let $0 < z < y < x$ be three real numbers such that $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in an arithmetic progression and $x, \sqrt{2}y, z$ are in a

geometric progression. If $xy + yz + zx = \frac{3}{\sqrt{2}}xyz$, then $3(x + y + z)^2$ is equal to _____.

माना तीन वास्तविक संख्याएँ $0 < z < y < x$ इस प्रकार हैं कि $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ एक समांतर श्रेढ़ी में हैं तथा $x, \sqrt{2}y, z$ एक गुोत्तर श्रेढ़ी में

हैं। यदि $xy + yz + zx = \frac{3}{\sqrt{2}}xyz$ है, तो $3(x + y + z)^2$ बराबर है _____

Ans. 150

Sol. As given, $\frac{2}{y} = \frac{1}{x} + \frac{1}{z}$ (i)

$$2y^2 = xz \quad \dots(ii)$$

$$\text{and } \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{3}{\sqrt{2}} \quad \dots(iii)$$

$$\text{From (i) and (iii) } y = \sqrt{2} \quad \dots(iv)$$

Now from (ii) $xz = 4$ (v)

Now using (ii), (iv) and (v)

$$x + z = 4\sqrt{2}$$

$$\begin{aligned}\text{Required, } 3(x + y + z)^2 &= 3(\sqrt{2} + 4\sqrt{2})^2 \\ &= 150\end{aligned}$$

86. Suppose $a_1, a_2, 2, a_3, a_4$ be in an arithmetico-geometric progression. If the common ratio of the corresponding geometric progression is 2 and the sum of all 5 terms of the arithmetico-geometric progression is $\frac{49}{2}$, then a_4 is equal to _____.

माना $a_1, a_2, 2, a_3, a_4$ एक समांतर-गुणोत्तर श्रेणी है। यदि संगत गुणोत्तर श्रेणी का सार्व अनुपात 2 है तथा समांतर-गुणोत्तर श्रेणी के सभी 5 पदों का योग $\frac{49}{2}$ है, तो a_4 बराबर है _____

Ans. 16

Sol. $\frac{(a-2d)}{4}, \frac{(a-d)}{2}, a, 2(a+d), 4(a+2d)$

$$a = 2$$

$$\left(\frac{1}{4} + \frac{1}{2} + 1 + 6\right) \times 2 + (-1 + 2 + 8)d = \frac{49}{2}$$

$$2\left(\frac{3}{4} + 7\right) + 9d = \frac{49}{2}$$

$$9d = \frac{49}{2} - \frac{62}{4} = \frac{98-62}{4} = 9$$

$$d = 1$$

$$\Rightarrow a_4 = 4(a+2d)$$

$$= 16$$

87. For $k \in \mathbb{N}$, if the sum of the series $1 + \frac{4}{k} + \frac{8}{k^2} + \frac{13}{k^3} + \frac{19}{k^4} + \dots$ is 10, then the value of k is _____.

$k \in \mathbb{N}$ के लिए, यदि श्रेणी $1 + \frac{4}{k} + \frac{8}{k^2} + \frac{13}{k^3} + \frac{19}{k^4} + \dots$ का योग 10 है, तो k का मान है _____

Ans. 2

Sol. $10 = 1 + \frac{4}{k} + \frac{8}{k^2} + \frac{13}{k^3} + \frac{19}{k^4} + \dots \text{upto } \infty$

$$9 = \frac{4}{k} + \frac{8}{k^2} + \frac{13}{k^3} + \frac{19}{k^4} + \dots \text{upto } \infty$$

$$\frac{9}{k} = \frac{4}{k^2} + \frac{8}{k^3} + \frac{13}{k^4} + \dots \text{ upto } \infty$$

$$S = 9\left(1 - \frac{1}{k}\right) = \frac{4}{k} + \frac{4}{k^2} + \frac{5}{k^3} + \frac{6}{k^4} + \dots \text{ upto } \infty$$

$$\frac{S}{k} = \frac{4}{k^2} + \frac{4}{k^3} + \frac{5}{k^4} + \dots \text{ upto } \infty$$

$$\left(1 - \frac{1}{k}\right)S = \frac{4}{k} + \frac{1}{k^3} + \frac{1}{k^4} + \frac{1}{k^5} + \dots \infty$$

$$9\left(1 - \frac{1}{k}\right)^2 = \frac{4}{k} + \frac{\frac{1}{k^3}}{\left(1 - \frac{1}{k}\right)}$$

$$9(k-1)^3 = 4k(k-1) + 1$$

$$k = 2$$

88. If the sum of the series

$$\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{2^2} - \frac{1}{2 \cdot 3} + \frac{1}{3^2}\right) + \left(\frac{1}{2^3} - \frac{1}{2^2 \cdot 3} + \frac{1}{2 \cdot 3^2} - \frac{1}{3^3}\right) + \left(\frac{1}{2^4} - \frac{1}{2^3 \cdot 3} + \frac{1}{2^2 \cdot 3^2} - \frac{1}{2 \cdot 3^3} + \frac{1}{3^4}\right) + \dots \text{ is } \frac{\alpha}{\beta},$$

where α and β are co-prime, then $\alpha + 3\beta$ is equal to _____.

यदि श्रेणी

$$\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{2^2} - \frac{1}{2 \cdot 3} + \frac{1}{3^2}\right) + \left(\frac{1}{2^3} - \frac{1}{2^2 \cdot 3} + \frac{1}{2 \cdot 3^2} - \frac{1}{3^3}\right) + \left(\frac{1}{2^4} - \frac{1}{2^3 \cdot 3} + \frac{1}{2^2 \cdot 3^2} - \frac{1}{2 \cdot 3^3} + \frac{1}{3^4}\right) + \dots \text{ का योग } \frac{\alpha}{\beta}$$

है, जहाँ α तथा β सहभाज्य है, तो $\alpha + 3\beta$ का मान होगा _____

Ans. 7

Sol. $P = \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{2^2} - \frac{1}{2 \cdot 3} + \frac{1}{3^2}\right) + \left(\frac{1}{2^3} - \frac{1}{2^2 \cdot 3} + \frac{1}{2 \cdot 3^2} - \frac{1}{3^3}\right) + \dots$

$$P\left(\frac{1}{2} + \frac{1}{3}\right) = \left(\frac{1}{2^2} - \frac{1}{3^2}\right) + \left(\frac{1}{2^3} + \frac{1}{3^3}\right) + \left(\frac{1}{2^4} - \frac{1}{3^4}\right) + \dots$$

$$\frac{5P}{6} = \frac{\frac{1}{4}}{1 - \frac{1}{2}} - \frac{\frac{1}{9}}{1 + \frac{1}{3}}$$

$$\frac{5P}{6} = \frac{1}{2} - \frac{1}{12} = \frac{5}{12}$$

$$\therefore P = \frac{1}{2} = \frac{\alpha}{\beta}$$

$$\therefore \alpha = 1, \beta = 2$$

$$\alpha + 3\beta = 7$$

Determinant

89. The number of values of $\theta \in (0, \pi)$ for which the system of linear equations

$$x + 3y + 7z = 0$$

$$-x + 4y + 7z = 0$$

$(\sin 3\theta)x + (\cos 2\theta)y + 2z = 0$ has a non-trivial solution, is:

(A) three

(B) four

(C) two

(D) one

$\theta \in (0, \pi)$ के ऐसे मानों की संख्या, जिनके लिए निम्न रैखिक समीकरण निकाय

$$x + 3y + 7z = 0$$

$$-x + 4y + 7z = 0$$

$(\sin 3\theta)x + (\cos 2\theta)y + 2z = 0$ का एक अतुच्छ हल है, है:

(A) तीन

(B) चार

(C) दो

(D) एक

Ans. C

Sol. for non trivial solution

$$\begin{vmatrix} 1 & 3 & 7 \\ -1 & 4 & 7 \\ \sin 3\theta & \cos 2\theta & 2 \end{vmatrix} = 0$$

$$\sin 3\theta(21 - 28) - \cos 2\theta(7 + 7) + 2(4 + 3) = 0$$

$$-7\sin 3\theta - 14\cos 2\theta + 14 = 0$$

$$-[3\sin \theta - 4\sin^3 \theta] - 2(1 - 2\sin^2 \theta) + 2 = 0$$

$$4\sin^3 \theta + 4\sin^2 \theta - 3\sin \theta = 0$$

$$\sin \theta [4\sin^2 \theta + 4\sin \theta - 3] = 0$$

$$\sin \theta = 0$$

$$2\sin \theta(2\sin \theta + 3) - (2\sin \theta + 3) = 0$$

$$(2\sin \theta - 1)(2\sin \theta + 3) = 0$$

$$\sin \theta = \frac{1}{2}$$

$$\sin \theta = \frac{-3}{2} \text{ Not possible}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

90. Let $a_1, a_2, a_3, \dots, a_{10}$ be in G.P. with $a_i > 0$ for $i = 1, 2, \dots, 10$ and S be the set of pairs (r, k) , $k \in \mathbb{N}$ (the

set of natural numbers) for which
$$\begin{vmatrix} \log_e a_1^r a_2^k & \log_e a_2^r a_3^k & \log_e a_3^r a_4^k \\ \log_e a_4^r a_5^k & \log_e a_5^r a_6^k & \log_e a_6^r a_7^k \\ \log_e a_7^r a_8^k & \log_e a_8^r a_9^k & \log_e a_9^r a_{10}^k \end{vmatrix} = 0$$

Then the number of elements in S, is:

- (A) 4 (B) 10 (C) 2 (D) Infinity many

माना $a_1, a_2, a_3, \dots, a_{10}$ गुणोत्तर श्रेणी में है तथा $i = 1, 2, \dots, 10$ के लिए $a_i > 0$ है और S ऐसे युग्मों $(r, k), k \in \mathbb{N}$

(प्राकृत संख्याओं का समुच्चय) का समुच्चय है जिनके लिए
$$\begin{vmatrix} \log_e a_1^r a_2^k & \log_e a_2^r a_3^k & \log_e a_3^r a_4^k \\ \log_e a_4^r a_5^k & \log_e a_5^r a_6^k & \log_e a_6^r a_7^k \\ \log_e a_7^r a_8^k & \log_e a_8^r a_9^k & \log_e a_9^r a_{10}^k \end{vmatrix} = 0$$

तो S के अवयवों की संख्या है

- (A) 4 (B) 10 (C) 2 (D) अनन्त

Ans. D

Sol. $\therefore a_1, a_2, \dots, a_{10} \rightarrow GP$
 $\therefore \ln a_1, \ln a_2, \ln a_3, \dots, \ln a_{10} \rightarrow AP$

$$\begin{vmatrix} r \ln a_1 + k \ln a_2 & r \ln a_2 + k \ln a_3 & r \ln a_3 + k \ln a_4 \\ r \ln a_4 + k \ln a_5 & r \ln a_5 + k \ln a_6 & r \ln a_6 + k \ln a_7 \\ r \ln a_7 + k \ln a_8 & r \ln a_8 + k \ln a_9 & r \ln a_9 + k \ln a_{10} \end{vmatrix} = 0$$

Apply $C_1 \rightarrow C_1 - C_2, C_2 \rightarrow C_2 - C_3$

Now, $C_1 = C_2$

So determinant = zero

91. If
$$\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)(x+a+b+c)^2, x \neq 0 \text{ and } a+b+c \neq 0, \text{ then } x \text{ is equal to}$$

- (A) abc (B) $-(a+b+c)$ (C) $-2(a+b+c)$ (D) $2(a+b+c)$

यदि
$$\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)(x+a+b+c)^2, x \neq 0 \text{ तथा } a+b+c \neq 0, \text{ तो } x \text{ बराबर है—}$$

- (A) abc (B) $-(a+b+c)$ (C) $-2(a+b+c)$ (D) $2(a+b+c)$

Ans. C

Sol. $R_1 \rightarrow R_1 + R_2 + R_3$

$$(a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$$c_1 \rightarrow c_1 \rightarrow c_2 \quad c_2 \rightarrow c_2 \rightarrow c_3$$

$$(a+b+c) \begin{vmatrix} 0 & 0 & 1 \\ a+b+c & -(a+b+c) & 2b \\ 0 & a+b+c & c-a-b \end{vmatrix}$$

$$(a+b+c)[1(a+b+c)]^2 = (a+b+c)(x+a+b+c)^2$$

$$x+a+b+c = \pm(a+b+c)$$

$$x = -2(a+b+c)$$

92. An ordered pair (α, β) for which the system of linear equations :

$$(1 + \alpha)x + \beta y + z = 2$$

$$\alpha x + (1 + \beta)y + z = 3$$

$$\alpha x + \beta y + 2z = 2$$

has a unique solution, is :

(A) $(-4, 2)$ (B) $(2, 4)$ (C) $(-3, 1)$ (D) $(1, -3)$

एक ऐसा क्रमित युग्म (α, β) जिसके लिए रेखिक समीकरण निकाय

$$(1 + \alpha)x + \beta y + z = 2$$

$$\alpha x + (1 + \beta)y + z = 3$$

$$\alpha x + \beta y + 2z = 2$$

का एकमात्र एक हल है, हैं -

(A) $(-4, 2)$ (B) $(2, 4)$ (C) $(-3, 1)$ (D) $(1, -3)$

Ans. B

Sol. For unique solution $\Delta \neq 0$

$$\begin{vmatrix} 1+\alpha & \beta & 1 \\ \alpha & 1+\beta & 1 \\ \alpha & \beta & 2 \end{vmatrix} \neq 0$$

$$\begin{vmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ \alpha & \beta & 2 \end{vmatrix} \neq 0$$

$$1(2+\beta) + 1(+\alpha) \neq 0$$

$$\alpha + \beta + 2 \neq 0$$

93. The system of linear equations :

$$x + y + z = 2$$

$$2x + 3y + 2z = 5$$

$$2x + 3y + (a^2 - 1)z = a + 1$$

(A) Is inconsistent when $|a| = \sqrt{3}$

(B) Has a unique solution for $|a| = \sqrt{3}$

(C) Has infinitely many solution for $a = 4$

(D) is inconsistent when $a = 4$

रैखिक समीकरण निकाय

$$x + y + z = 2$$

$$2x + 3y + 2z = 5$$

$$2x + 3y + (a^2 - 1)z = a + 1$$

(A) असंगत है जब $|a| = \sqrt{3}$

(B) का $|a| = \sqrt{3}$ के लिए अद्वितीय हल है।

(C) के $a = 4$ के लिए अनन्त हल हैं।

(D) असंगत है जब $a = 4$

Ans. A

Sol.
$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 2 & 3 & (a^2 - 1) \end{vmatrix} = 0$$

$$\Rightarrow |a| = \sqrt{3}$$

$\Delta_x \neq 0$ for $a = \sqrt{3}$ so system of equations are inconsistent

94. If the system of equations $2x + 3y - z = 0$, $x + ky - 2z = 0$ and $2x - y + z = 0$ has a non-trivial solution (x, y, z) , then

$\frac{x}{y} + \frac{y}{z} + \frac{z}{x} + k$ is equal to:

(A) $\frac{1}{2}$

(B) -4

(C) $\frac{3}{4}$

(D) $-\frac{1}{4}$

यदि समीकरण निकाय $2x + 3y - z = 0$, $x + ky - 2z = 0$ तथा $2x - y + z = 0$ का एक अतुच्छ (non-trivial) हल (x, y, z) है,

तो $\frac{x}{y} + \frac{y}{z} + \frac{z}{x} + k$ बराबर है :

(A) $\frac{1}{2}$

(B) -4

(C) $\frac{3}{4}$

(D) $-\frac{1}{4}$

Ans. A

Sol.
$$\Delta = 0 \Rightarrow \begin{vmatrix} 2 & 3 & -1 \\ 1 & k & -2 \\ 2 & -1 & 1 \end{vmatrix} = 0 \Rightarrow \boxed{k = \frac{9}{2}}$$

$\therefore$ Equations are $2x + 3y - z = 0$ (i)

$2x - y + z = 0$ (ii)

$$2x + 9y - 4z = 0 \quad \dots(\text{iii})$$

By (i) – (ii) $\boxed{2y = z} \therefore \boxed{z = -4x}$ and $\boxed{2x + y = 0}$

$$\therefore \frac{x}{y} + \frac{y}{z} + \frac{z}{x} + k = \frac{-1}{2} + \frac{1}{2} - 4 + \frac{9}{2}$$

$$= \frac{1}{2}$$

95. If $\Delta_1 = \begin{vmatrix} x & \sin \theta & \cos \theta \\ -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & \sin 2\theta & \cos 2\theta \\ -\sin 2\theta & -x & 1 \\ \cos 2\theta & 1 & x \end{vmatrix}$, $x \neq 0$; then for all $\theta \in \left(0, \frac{\pi}{2}\right)$:

(A) $\Delta_1 + \Delta_2 = -2x^3$

(B) $\Delta_1 - \Delta_2 = -2x^3$

(C) $\Delta_1 + \Delta_2 = -2(x^3 + x - 1)$

(D) $\Delta_1 - \Delta_2 = x(\cos 2\theta - \cos 4\theta)$

यदि $\Delta_1 = \begin{vmatrix} x & \sin \theta & \cos \theta \\ -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix}$ तथा $\Delta_2 = \begin{vmatrix} x & \sin 2\theta & \cos 2\theta \\ -\sin 2\theta & -x & 1 \\ \cos 2\theta & 1 & x \end{vmatrix}$, $x \neq 0$; तो सभी $\theta \in \left(0, \frac{\pi}{2}\right)$ के लिए :

(A) $\Delta_1 + \Delta_2 = -2x^3$

(B) $\Delta_1 - \Delta_2 = -2x^3$

(C) $\Delta_1 + \Delta_2 = -2(x^3 + x - 1)$

(D) $\Delta_1 - \Delta_2 = x(\cos 2\theta - \cos 4\theta)$

Ans. A

Sol. $\Delta_1 = \begin{vmatrix} x & \sin \theta & \cos \theta \\ -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix}$

$$= x(-x^2 - 1) - \sin \theta(-x \sin \theta - \cos \theta) + \cos \theta(-\sin \theta + x \cos \theta)$$

$$= -x^3 - x + x \sin^2 \theta + \sin \theta \cos \theta - \cos \theta \sin \theta + x \cos^2 \theta$$

$$= -x^3 - x + x = -x^3$$

Similarly $\Delta_2 = -x^3$

$$\Delta_1 + \Delta_2 = -2x^3$$

96. The sum of the real roots of the equation $\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$, is equal to :

(A) 6

(B) 0

(C) -4

(D) 1

समीकरण $\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$, के वास्तविक मूलों का योगफल है :

- (A) 6 (B) 0 (C) -4 (D) 1

Ans. B

Sol. $\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$

$$\Rightarrow x(-3x^2 - 6x - 2x^2 + 6x) - 6(-3x + 9 - 2x - 4) - (4x - 9x) = 0$$

$$\Rightarrow x(-5x^2) - 6(-5x + 5) - 4x + 9x = 0$$

$$\Rightarrow x^3 - 7x + 6 = 0$$

All the roots are real

$$\therefore \text{Sum of real roots} = \frac{0}{1} = 0$$

97. The system of linear equations

$$\lambda x + 2y + 2z = 5$$

$$2\lambda x + 3y + 5z = 8$$

$$4x + \lambda y + 6z = 10 \quad \text{has :}$$

- (A) a unique solution when $\lambda = -8$ (B) no solution when $\lambda = 2$
(C) no solution when $\lambda = 8$ (D) infinitely many solutions when $\lambda = 2$

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$$\lambda x + 2y + 2z = 5$$

$$2\lambda x + 3y + 5z = 8$$

$$4x + \lambda y + 6z = 10 \quad \text{है :}$$

- (A) का मात्र एक हल है जब $\lambda = -8$ (B) का कोई हल नहीं है जब $\lambda = 2$
(C) का कोई हल नहीं है जब $\lambda = 8$ (D) के अनन्त हल हैं जब $\lambda = 2$

Ans. B

Sol. $D = \begin{vmatrix} \lambda & 2 & 2 \\ 2\lambda & 3 & 5 \\ 4 & \lambda & 6 \end{vmatrix} = (\lambda + 8)(2 - \lambda)$

For unique solution $\lambda \neq 2, -8$
for $\lambda = 2$

$$D_x = \begin{vmatrix} 5 & 2 & 2 \\ 8 & 3 & 5 \\ 10 & 2 & 6 \end{vmatrix}$$

$$D_x = 16 \neq 0 \quad (\text{No solution})$$

98. If for some α and β in $\mathbb{R}$, the intersection of the following three planes

$$x + 4y - 2z = 1$$

$$x + 7y - 5z = \beta$$

$$x + 5y + \alpha z = 5$$

is a line in $\mathbb{R}^3$, then $\alpha + \beta$ is equal to :

(A) 2

(B) -10

(C) 0

(D) 10

यदि $\mathbb{R}$ में किन्हीं α तथा β के लिए निम्न तीन समतलों

$$x + 4y - 2z = 1$$

$$x + 7y - 5z = \beta$$

$$x + 5y + \alpha z = 5$$

का प्रतिच्छेदन, $\mathbb{R}^3$ में एक रेखा है, तो $\alpha + \beta$ का मान है :

(A) 2

(B) -10

(C) 0

(D) 10

Ans. D

Sol. $x + 4y - 2z = 1$

$$x + 7y - 5z = \beta$$

$$x + 5y + \alpha z = 5$$

$\Rightarrow$

System has infinite solutions

$$D = \begin{vmatrix} 1 & 4 & -2 \\ 1 & 7 & -5 \\ 1 & 5 & \alpha \end{vmatrix} = 0$$

$\Rightarrow$

$$1(7\alpha + 25) - 4(\alpha + 5) - 2(5 - 7) = 0$$

$$7\alpha + 25 - 4\alpha - 20 + 4 = 0$$

$$3\alpha + 9 = 0$$

$$\therefore \alpha = -3$$

$$D_x = \begin{vmatrix} 1 & 4 & -2 \\ \beta & 7 & -5 \\ 5 & 5 & -3 \end{vmatrix} = 0$$

$$\Rightarrow 1(-21 + 25) - 4(-3\beta + 25) - 2(5\beta - 35) = 0$$

$$\Rightarrow 4 + 12\beta - 100 - 10\beta + 70 = 0$$

$$2\beta - 26 = 0$$

$$\beta = 13$$

$$D_y = \begin{vmatrix} 1 & 1 & -2 \\ 1 & \beta & -5 \\ 1 & 5 & -3 \end{vmatrix} = 0$$

$$\Rightarrow 1(-3\beta + 25) - 1(-3 + 10) + 1(-5 + 2\beta) = 0$$

$$-3\beta + 25 - 7 - 5 + 2\beta = 0$$

$$-\beta + 13 = 0$$

$$\therefore \beta = 13$$

$$D_z = \begin{vmatrix} 1 & 4 & 1 \\ 1 & 7 & \beta \\ 1 & 5 & 5 \end{vmatrix} = 0$$

$$\Rightarrow 1(5 - 7) - \beta(5 - 4) + 5(7 - 4) = 0$$

$$-2 - \beta + 15 = 0$$

$$\beta = 13$$

$$\alpha + \beta = -3 + 13 = 10$$

99. Let $a - 2b + c = 1$.

$$\text{If } f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}, \text{ then :}$$

(A) $f(50) = 1$ (B) $f(50) = -501$ (C) $f(-50) = -1$ (D) $f(-50) = 501$

माना Let $a - 2b + c = 1$ है | यदि

$$\text{If } f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix} \text{ है, तो :}$$

(A) $f(50) = 1$ (B) $f(50) = -501$ (C) $f(-50) = -1$ (D) $f(-50) = 501$

Ans. A

Sol. $a - 2b + c = 1$

$$f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_3 - 2R_2$$

$$f(x) = \begin{vmatrix} 2x + a + c - 2x - 2b & 2x + 6 - 2x - 6 & 2x + 4 - 2x - 4 \\ x + b & x + 3 & x + 2 \\ x + c & x + 4 & x + 3 \end{vmatrix}$$

$$f(x) = \begin{vmatrix} 1 & 0 & 0 \\ x + b & x + 3 & x + 2 \\ x + c & x + 4 & x + 3 \end{vmatrix}$$

$$= 1((x+3)^2 - (x+4)(x+2))$$

$$= x^2 + 6x + 9 - x^2 - 6x - 8 = 1$$

100. The following system of linear equations

$$7x + 6y - 2z = 0$$

$$3x + 4y + 2z = 0$$

$$x - 2y - 6z = 0, \text{ has}$$

(A) only the trivial solution

(B) infinitely many solutions, (x, y, z) satisfying $x = 2z$

(C) infinitely many solutions, (x, y, z) satisfying $y = 2z$

(D) no solution

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$$7x + 6y - 2z = 0$$

$$3x + 4y + 2z = 0$$

$$x - 2y - 6z = 0$$

(A) का केवल तुच्छ हल हैं।

(B) $x = 2z$ को सन्तुष्ट करने वाले अनन्त हल (x, y, z) हैं।

(C) $y = 2z$ को सन्तुष्ट करने वाले अनन्त हल (x, y, z) हैं।

(D) का कोई हल नहीं है।

Ans. B

Sol. $7x + 6y - 2z = 0$

$$3x + 4y + 2z = 0$$

$$x - 2y - 6z = 0$$

$$D = \begin{vmatrix} 7 & 6 & -2 \\ 3 & 4 & 2 \\ 1 & -2 & -6 \end{vmatrix} = 7(-24 + 4) - 6(-18 - 2) - 2(-6 - 4)$$

$$= 7(-20) - 6(-20) - 2(-10)$$

$$= -140 + 120 + 20 = 0$$

$$7x + 6y = 2t$$

$$3x + 4y = -2t$$

$$21x + 18y = 6t$$

$$21x + 28y = -14t$$

$$\begin{array}{r} 21x + 18y = 6t \\ 21x + 28y = -14t \\ \hline -10y = 20t \end{array} \Rightarrow y = -2t$$

$$7x + 6(-2t) = 2t$$

$$7x = 14t \Rightarrow x = 2t$$

$$x = 2z$$